

Practice Sheet: FILTER Function

This practice sheet helps you master the **FILTER** function in Excel. The FILTER function is used to extract records from a table based on one or more conditions.

It is very useful when you want to show only selected records, such as only DCA students, only active students, students with marks above 80, students from a particular batch, and so on.

New Function: FILTER

=FILTER(array, include, [if_empty])

Meaning:

- **array**: The full range of data that you want to filter.
- **include**: The condition that decides which rows should be shown.
- **if_empty**: The message to show if no result is found.

Example:

=FILTER(A2:G51,C2:C51="DCA","No Records Found")

This formula displays only the students whose course is **DCA**.

Important Note: The FILTER function is available in newer versions of Excel such as **Excel 2021** and Microsoft 365.

Step 1: Set Up the Main Data Table

In Sheet1, starting from cell **A1**, enter the following table.

Roll No.	Student Name	Course	Batch	Marks	Attendance %	Status
S001	Aarav Sharma	DCA	Morning	78	92	Active
S002	Bhavna Patel	PGDCA	Evening	45	75	Active
S003	Chirag Reddy	ADCA	Morning	89	98	Active
S004	Diya Singh	DCA	Afternoon	62	88	Active
S005	Eshan Kumar	PGDCA	Morning	55	70	Inactive
S006	Fiza Khan	DCA	Evening	91	96	Active
S007	Gaurav Joshi	ADCA	Afternoon	72	82	Active
S008	Harini Nair	DCA	Morning	38	65	Inactive
S009	Ishaan Rao	PGDCA	Evening	84	94	Active
S010	Jaya Menon	ADCA	Morning	67	85	Active
S011	Kiran Desai	DCA	Afternoon	93	97	Active
S012	Lata Gupta	PGDCA	Morning	50	78	Active
S013	Mohan Rao	ADCA	Evening	76	90	Active

Roll No.	Student Name	Course	Batch	Marks	Attendance %	Status
S014	Nidhi Sharma	DCA	Morning	61	81	Active
S015	Ojas Yadav	PGDCA	Afternoon	44	68	Inactive
S016	Priya Thakur	ADCA	Evening	88	95	Active
S017	Qasim Ali	DCA	Morning	73	89	Active
S018	Riya Choudhary	PGDCA	Evening	81	93	Active
S019	Sohan Malhotra	ADCA	Afternoon	69	87	Active
S020	Tanya Saxena	DCA	Morning	56	79	Active
S021	Udit Narayan	PGDCA	Evening	95	99	Active
S022	Vani Mishra	ADCA	Morning	64	83	Active
S023	Waman Dubey	DCA	Afternoon	82	91	Active
S024	Xena Grover	PGDCA	Morning	51	76	Active
S025	Yashwant Kaur	ADCA	Evening	77	88	Active
S026	Zara Vardhan	DCA	Morning	85	96	Active
S027	Aarti Jain	PGDCA	Afternoon	59	80	Active
S028	Bharat Sethi	ADCA	Evening	70	85	Active
S029	Chanda Bose	DCA	Morning	48	72	Inactive
S030	Dheeraj Menon	PGDCA	Afternoon	86	94	Active
S031	Esha Agarwal	ADCA	Evening	74	90	Active
S032	Farhan Pandey	DCA	Morning	92	98	Active
S033	Gitanjali Mishra	PGDCA	Evening	63	84	Active
S034	Hitesh Saxena	ADCA	Afternoon	80	92	Active
S035	Indira Grover	DCA	Morning	54	77	Active
S036	Jagdish Kaur	PGDCA	Evening	87	95	Active
S037	Kavya Vardhan	ADCA	Morning	71	88	Active
S038	Lalit Jain	DCA	Afternoon	49	74	Inactive
S039	Manisha Sethi	PGDCA	Evening	83	93	Active
S040	Naveen Bose	ADCA	Morning	66	86	Active
S041	Ojaswi Menon	DCA	Afternoon	94	99	Active
S042	Parth Agarwal	PGDCA	Morning	57	81	Active

Roll No.	Student Name	Course	Batch	Marks	Attendance %	Status
S043	Qurat Pandey	ADCA	Evening	79	91	Active
S044	Roshni Mishra	DCA	Morning	52	75	Active
S045	Siddhant Saxena	PGDCA	Afternoon	88	96	Active
S046	Tanvi Grover	ADCA	Evening	65	83	Active
S047	Umesh Kaur	DCA	Morning	96	100	Active
S048	Vikash Vardhan	PGDCA	Afternoon	60	80	Active
S049	Yogesh Bora	ADCA	Evening	82	94	Active
S050	Zainab Das	DCA	Morning	73	89	Active

Step 2: Filter Only DCA Students

In cell **I1**, type:

DCA Students

In cell **I2**, enter the following formula:

`=FILTER(A2:G51,C2:C51="DCA","No Records Found")`

This will display only the students whose course is **DCA**.

Step 3: Filter Only PGDCA Students

In cell **Q1**, type:

PGDCA Students

In cell **Q2**, enter the following formula:

`=FILTER(A2:G51,C2:C51="PGDCA","No Records Found")`

This will display only the students whose course is **PGDCA**.

Step 4: Filter Only ADCA Students

In cell **Y1**, type:

ADCA Students

In cell **Y2**, enter the following formula:

`=FILTER(A2:G51,C2:C51="ADCA","No Records Found")`

This will display only the students whose course is **ADCA**.

Step 5: Filter Students Who Scored More Than 80

In cell **I25**, type:

Marks Above 80

In cell **I26**, enter the following formula:

`=FILTER(A2:G51,E2:E51>80,"No Records Found")`

This will display only those students whose marks are more than **80**.

Step 6: Filter Students Who Scored Less Than 60

In cell **Q25**, type:

Marks Below 60

In cell **Q26**, enter the following formula:

`=FILTER(A2:G51,E2:E51<60,"No Records Found")`

This will display only those students whose marks are below **60**.

Step 7: Filter Students With Attendance Less Than 75%

In cell **Y25**, type:

Attendance Below 75

In cell **Y26**, enter the following formula:

`=FILTER(A2:G51,F2:F51<75,"No Records Found")`

This will display only those students whose attendance is below **75%**.

Step 8: Filter Only Active Students

In cell **I50**, type:

Active Students

In cell **I51**, enter the following formula:

`=FILTER(A2:G51,G2:G51="Active","No Records Found")`

This will display only the students whose status is **Active**.

Step 9: Filter Only Inactive Students

In cell **Q50**, type:

Inactive Students

In cell **Q51**, enter the following formula:

`=FILTER(A2:G51,G2:G51="Inactive","No Records Found")`

This will display only the students whose status is **Inactive**.

Step 10: Filter Morning Batch Students

In cell **Y50**, type:

Morning Batch

In cell **Y51**, enter the following formula:

```
=FILTER(A2:G51,D2:D51="Morning","No Records Found")
```

This will display only the students from the **Morning** batch.

Step 11: Filter Evening Batch Students

In cell **I80**, type:

Evening Batch

In cell **I81**, enter the following formula:

```
=FILTER(A2:G51,D2:D51="Evening","No Records Found")
```

This will display only the students from the **Evening** batch.

Step 12: Filter Afternoon Batch Students

In cell **Q80**, type:

Afternoon Batch

In cell **Q81**, enter the following formula:

```
=FILTER(A2:G51,D2:D51="Afternoon","No Records Found")
```

This will display only the students from the **Afternoon** batch.

Step 13: Filter With Two Conditions

To filter students who are in **DCA** and scored more than **70**:

In cell **Y80**, type:

DCA Marks Above 70

In cell **Y81**, enter the following formula:

```
=FILTER(A2:G51,(C2:C51="DCA")*(E2:E51>70),"No Records Found")
```

Explanation:

- **C2:C51="DCA"** checks whether the course is DCA.
- **E2:E51>70** checks whether marks are above 70.
- The ***** symbol works like **AND**.

Step 14: Filter Active Students With Marks Above 80

In cell **I105**, type:

Active Marks Above 80

In cell **I106**, enter the following formula:

`=FILTER(A2:G51,(G2:G51="Active")*(E2:E51>80),"No Records Found")`

This will show students who are both **Active** and scored more than **80** marks.

Step 15: Filter Students With High Marks and High Attendance

In cell **Q105**, type:

High Marks and Attendance

In cell **Q106**, enter the following formula:

`=FILTER(A2:G51,(E2:E51>80)*(F2:F51>90),"No Records Found")`

This will show students who scored more than **80** and have attendance more than **90%**.

Step 16: Filter DCA or PGDCA Students

To filter students who are either in **DCA** or **PGDCA**:

In cell **Y105**, type:

DCA or PGDCA

In cell **Y106**, enter the following formula:

`=FILTER(A2:G51,(C2:C51="DCA")+(C2:C51="PGDCA"),"No Records Found")`

Explanation:

- The + symbol works like **OR**.
- This means the result will include students from either DCA or PGDCA.

Step 17: Filter Morning or Afternoon Batch Students

In cell **I135**, type:

Morning or Afternoon

In cell **I136**, enter the following formula:

`=FILTER(A2:G51,(D2:D51="Morning")+(D2:D51="Afternoon"),"No Records Found")`

This will show students from either the **Morning** batch or the **Afternoon** batch.

Step 18: Filter Using a Criteria Cell

Create the following criteria table in cells **Q135:R136**:

Cell	Value to Enter
Q135	Criteria
R135	Value
Q136	Course
R136	DCA

In cell **Q138**, type:

Filter by Course

In cell **Q139**, enter the following formula:

=FILTER(A2:G51,C2:C51=R136,"No Records Found")

Now change the value in cell **R136** to **PGDCA** or **ADCA** and observe how the result changes automatically.

Step 19: Filter Using Two Criteria Cells

Create the following criteria table in cells **Y135:Z137**:

Cell	Value to Enter
Y135	Criteria
Z135	Value
Y136	Course
Z136	DCA
Y137	Minimum Marks
Z137	70

In cell **Y139**, type:

Filter by Course and Marks

In cell **Y140**, enter the following formula:

=FILTER(A2:G51,(C2:C51=Z136)*(E2:E51>=Z137),"No Records Found")

Now change the course and minimum marks values and observe the result.

Step 20: Filter Using Three Criteria Cells

Create the following criteria table in cells **I165:J168**:

Cell	Value to Enter
I165	Criteria
J165	Value
I166	Course
J166	DCA
I167	Batch
J167	Morning
I168	Minimum Marks
J168	70

In cell **I170**, type:

Dynamic Filter

In cell **I171**, enter the following formula:

=FILTER(A2:G51,(C2:C51=J166)*(D2:D51=J167)*(E2:E51>=J168),"No Records Found")

Now change the criteria values and observe how the filtered result changes.

Practice Questions

Use the **FILTER** function to solve the following questions. Use the exact cells given below so that everyone's workbook remains uniform.

Cell	Task
A200	Show only the students from the DCA course.
A225	Show only the students from the PGDCA course.
A250	Show only the students from the ADCA course.
A275	Show only the students from the Morning batch.
A300	Show only the students from the Afternoon batch.
A325	Show only the students from the Evening batch.
A350	Show only the students who scored more than 90 .
A375	Show only the students who scored less than 50 .
A400	Show only the students whose attendance is more than 95% .
A425	Show only the students whose attendance is less than 75% .
A450	Show only the students whose status is Active .
A500	Show only the students whose status is Inactive .
A525	Show only the students who are in DCA and scored more than 80 .
A550	Show only the students who are in PGDCA and scored more than 75 .

Cell	Task
A575	Show only the students who are in ADCA and have attendance more than 90% .
A600	Show only the students who are Active and scored more than 85 .
A625	Show only the students who are Inactive and have attendance below 75% .
A650	Show only the students who are in the Morning batch and scored more than 70 .
A675	Show only the students who are in the Evening batch and have attendance more than 90% .
A700	Show only the students who are in DCA or ADCA .
A750	Show only the students who are in PGDCA or ADCA .
A800	Show only the students who are in the Morning or Evening batch.
A850	Show only the students who scored more than 80 and have attendance more than 90% .
A875	Show only the students who scored less than 60 or whose attendance is less than 75% .
A900	Show only the students who are Active , in DCA , and scored more than 70 .

Extra Challenge

Create a criteria table using **Course**, **Batch**, **Minimum Marks**, and **Status**.

Enter the following criteria table in cells **I930:J934**:

Cell	Value to Enter
I930	Criteria
J930	Value
I931	Course
J931	DCA
I932	Batch
J932	Morning
I933	Minimum Marks
J933	70
I934	Status
J934	Active

In cell **I936**, type:

Advanced Dynamic Filter

In cell **I937**, enter the following formula:

```
=FILTER(A2:G51,(C2:C51=J931)*(D2:D51=J932)*(E2:E51>=J933)*(G2:G51=J934),"No Records Found")
```

Now change the values in cells **J931:J934** and observe how the result changes.

Homework

Create a new sheet named **Homework**. In the Homework sheet, create a new table of **50 employees** starting from cell **A1** with the following columns:

Emp ID	Name	Department	Shift	Salary	Status
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Then use the **FILTER** function to find the following. Use the exact cells given below in the **Homework** sheet.

Cell	Task
H1	All employees from the Accounts department.
H25	All employees from the Sales department.
H50	All employees from the Computer department.
H75	Employees from the Morning shift.
H100	Employees from the Evening shift.
H125	Employees whose salary is above 25000 .
H150	Employees whose salary is below 15000 .
H175	Only Active employees.
H225	Only Inactive employees.
H250	Active employees with salary above 20000 .
H275	Accounts department employees with salary above 25000 .
H300	Sales or Accounts department employees.
H350	Morning or Evening shift employees.
H400	Inactive employees with salary below 18000 .
H425	Employees from Computer department who are Active.

Save your workbook as **Excel_Practice_Module14.xlsx** and submit it to your instructor.

End of Module 14